

Visium HD 3' Spatial Gene Expression

Site Preparation and Workflow Introduction

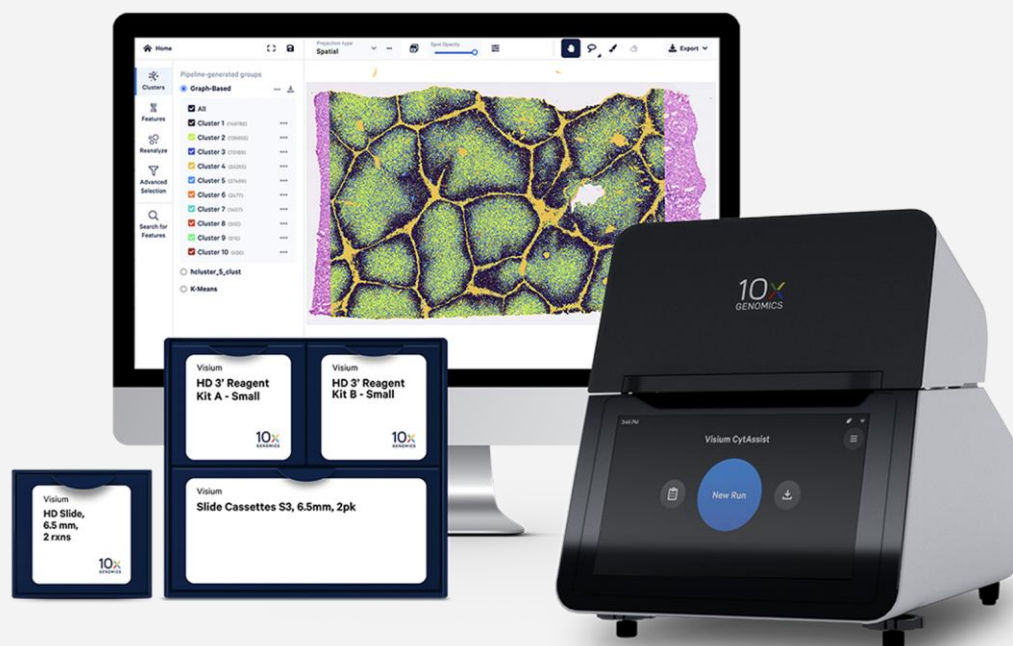
Peter Du 杜柏濤
Field Applications Scientist

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techsupport@gtbiotech.com.tw

Visium HD 3' Spatial Gene Expression

Product introduction

Visium HD 3' Spatial Gene Expression



Unbiased whole transcriptome analysis: 3' poly-A RNA capture

Single-cell scale: Continuous 6.5 mm grid of 2 μ m barcoded squares

Broad access: Species agnostic

Built for flexibility: Compatible with Fresh Frozen OCT embedded tissue sections on 25- or 26-mm wide slides

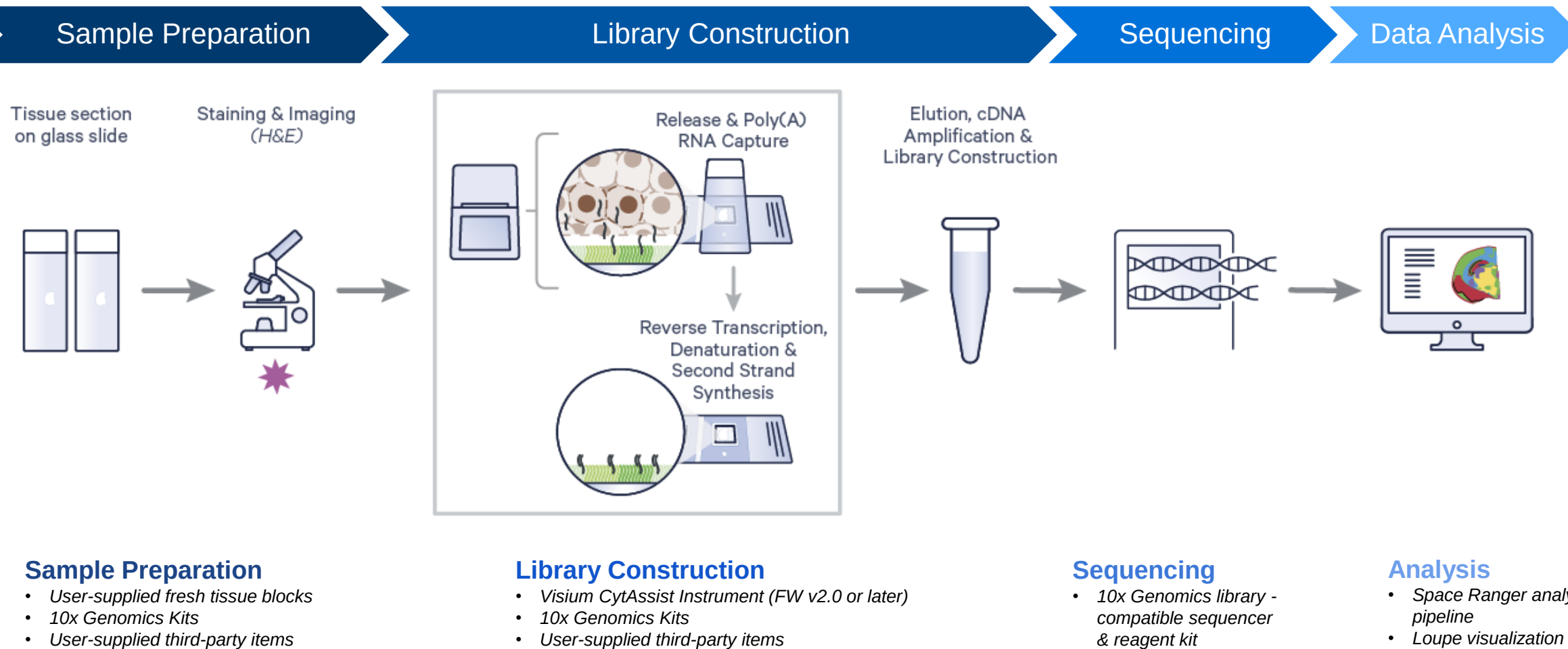
Enhanced histology: H&E staining from the same section

Consistent high-quality data: Powered by Visium CytAssist instrument running Firmware 2.0.0 or higher

Powerful analysis and visualization: With Space Ranger and Loupe Browser

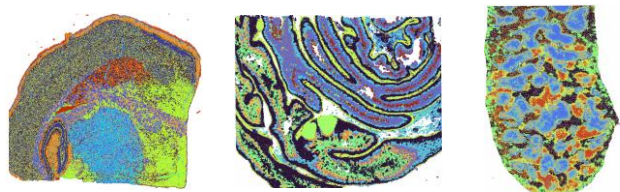
Visium HD 3' Spatial Gene Expression

Workflow overview



Visium HD 3' is compatible with a wide variety of tissue types and species

Mouse



Brain

Small intestine

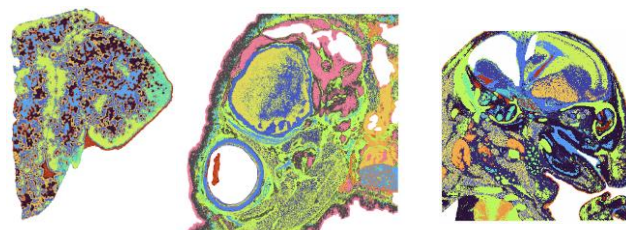
Spleen



Kidney

Lung

Heart

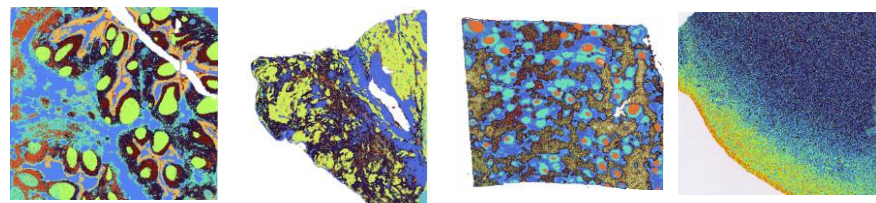


Thymus

Pup

Embryo

Human

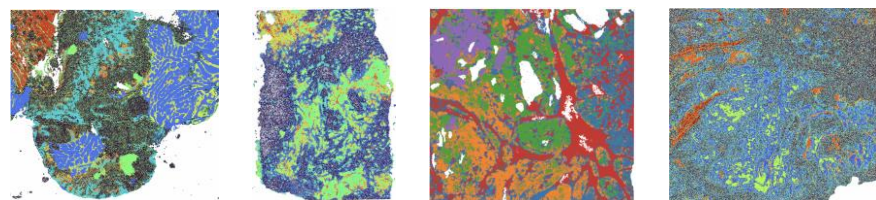


Tonsil

Ovarian cancer

Lymph node

Brain

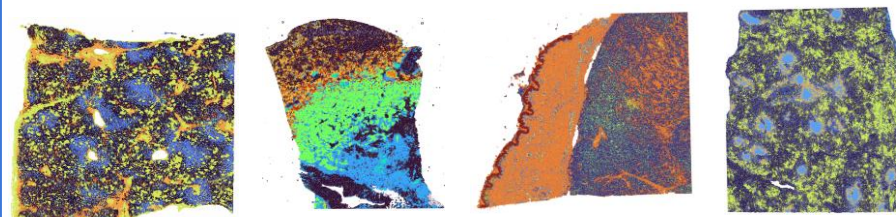


Pancreatic cancer

Breast cancer

Liver cancer

Colon cancer



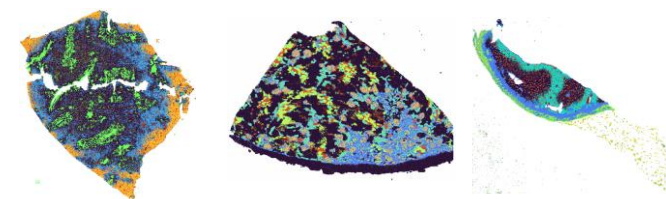
Liver

Lung cancer

Skin cancer

Spleen

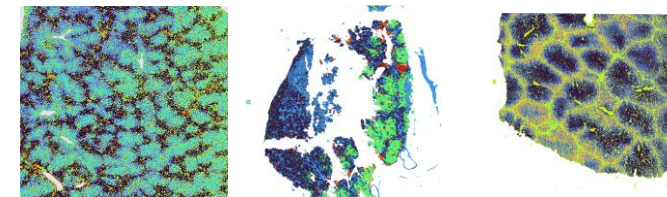
Other



Rhesus liver

Rhesus kidney

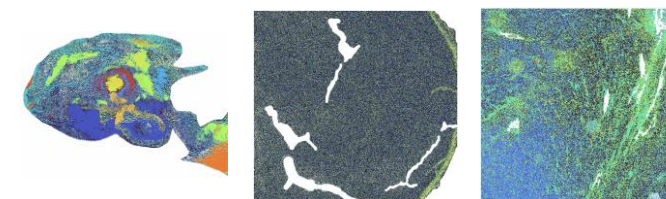
Cyno liver



Rat liver

Rat kidney

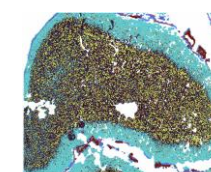
Pig liver



Zebrafish head

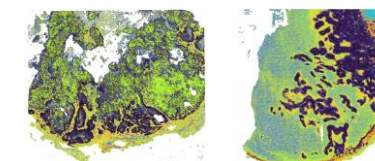
Chicken ovary

Chicken heart



Canine brain

Multi



Rat/human xenograft

Mouse/human xenograft

10x Genomics Kits & Storage

Visium HD 3' Spatial Gene Expression

Components, shipping and storage temperatures

Top Fill PN	Top Fill Name	Kit PN(s)	Kit Quantity	Components	Shipping Temp	Storage Temp
1000857	Visium HD 3', 6.5 mm, 4 rxn	1000670	2	Visium HD Slide, 6.5 mm, 2 rxns	Dry Ice	-80°C
		1000854	1	Visium HD 3' Reagents Kit A, Small		-20°C
		1000855	1	Visium HD 3' Reagents Kit B, Small		
		1000847	1	Visium Slide Cassettes S3, 6.5 mm, 2 pk	Ambient	RT
1000856	Visium HD 3', 6.5 mm, 16 rxn	1000670	8	Visium HD Slide, 6.5 mm, 2 rxns	Dry Ice	-80°C
		1000854	4	Visium HD 3' Reagents Kit A, Small		-20°C
		1000855	4	Visium HD 3' Reagents Kit B, Small		
		1000847	4	Visium Slide Cassettes S3 , 6.5 mm, 2 pk	Ambient	RT
1000215	Dual Index Kit TT Set A, 96 rxn	3000511	1	Dual Index Kit TT Set A	Dry Ice	-20°C
1000499	Visium CytAssist Reagent Accessory Kit	1000499	1	•1 Magnetic Separator (8 or 32 wells) •2 Low Profile Thermocycler Adapters	Ambient	RT
1000886	<i>Visium CytAssist Alignment Aid, 6.5 mm</i>	3002814	1	<i>Visium CytAssist Alignment Aid, 6.5 mm</i>	Ambient	RT
1000854	<i>Visium HD 3' Reagents Kit A, Small</i>	1000854	1	<i>Visium HD 3' Reagents Kit A, Small</i>	Dry Ice	-20°C

Optional

Visium HD 3' Spatial Gene Expression

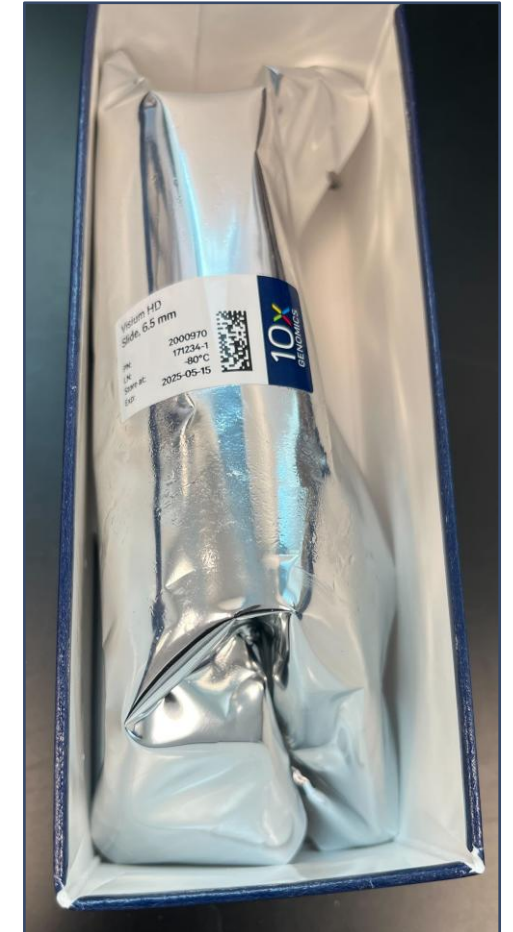
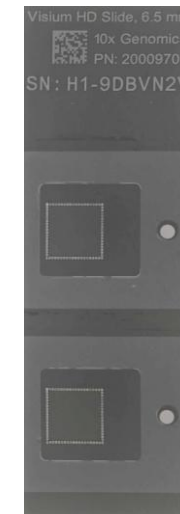
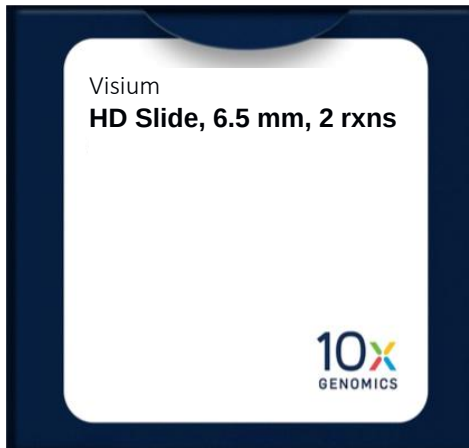
Required kits – Visium HD Slide Kit



- -80°C storage is required
- DO NOT open the mylar bag until slide is ready to be used

Visium HD Slide Kit, 6.5 mm, 2 rxns (PN-1000670)

- Visium HD Slide (1)
- Shipped on dry ice
- Store at -80°C



Visium HD 3' Spatial Gene Expression

Required kits – Visium HD Cassette Kit

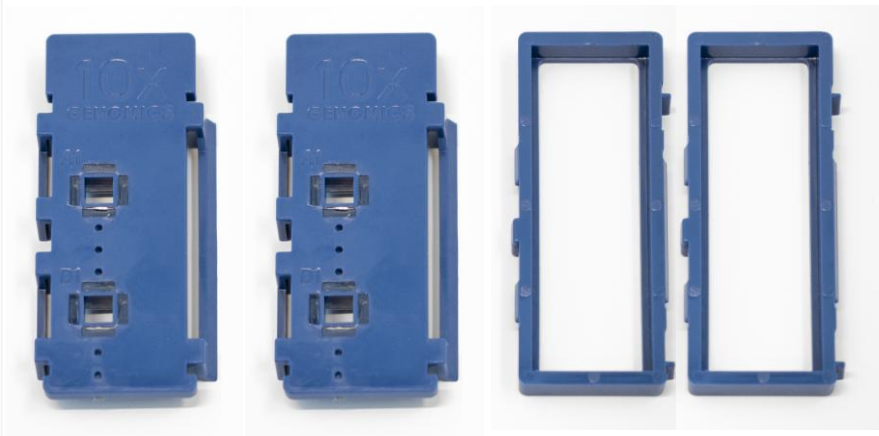
Visium Slide Cassettes S3, 6.5 mm, 2 pk (PN-1000847)

- Visium Cassette 2-port, 6.5 mm (2)
- Visium Slide Seals (1 x 12 pk)
- Shipped at ambient
- Store at ambient

Visium
Slide Cassettes S3, 6.5 mm, 2 pk



Visium 2-port Cassette S3



Visium Slide Seals



Visium HD 3' Spatial Gene Expression

- Shipping with **original Visium S3 2-port cassettes**, expect to ship **new** version later.
- Both cassette versions & part numbers are provided in documentation.

Visium Slide Cassettes S3*, 6.5 mm, 2 pk PN-1000847

Visium Slide Cassettes S3, 6.5 mm 2 pk PN-1000847 (store at ambient temperature)			
	#	PN	
Visium 2-port, 6.5 mm, 2 pk	1	2001252 or 2001344	3002329 or 3002328
Visium Cassette Bottom, 2 pk	1	2001344 or 2000283	3002328
Visium Slide Seals, 12 pack	1	2000283	

*The Visium HD 3' Workflow is run with Visium Cassettes S3. These are referred to as Visium Cassettes in this document. Consult the Visium Cassette S3 Quick Reference Card (CG000730) for assembly and disassembly information.

ORIGINAL

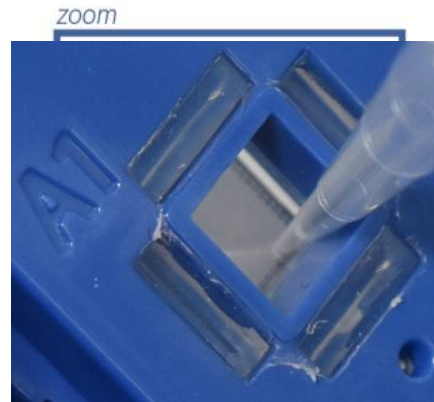
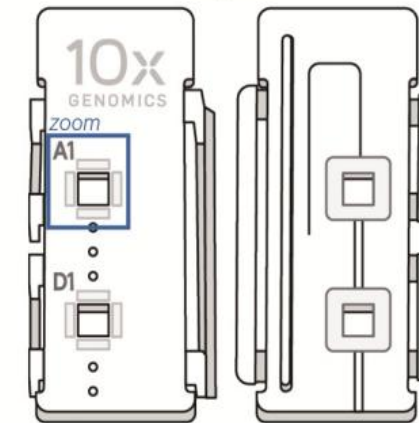


NEW

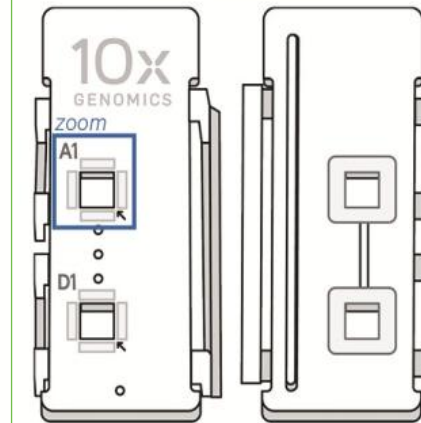


Variations in Visium HD Cassette Features*

Original



New Features



*All Visium HD Cassettes have the same functionality and performance.

ture Area is
rger towards
tom right
allow space
ing without
the hydrogel.
w points to
ct corner for
the pipette.

Visium HD 3' Spatial Gene Expression

Required kits – Reagent Kits

Visium HD 3' Reagent Kit A, Small (4 rxns) (PN-1000854)

- Contains reagents utilized during sample preparation & user guide steps
- Shipped on dry ice
- Store at -20°C



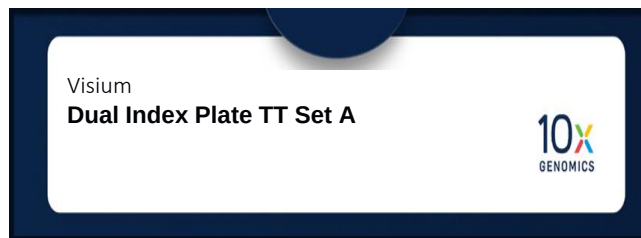
Visium HD 3' Reagent Kit B, Small (4 rxns) (PN-1000855)

- Contains reagents utilized during user guide steps
- Shipped on dry ice
- Store at -20°C



Dual Index Kit TT Set A, 96 rxns (PN-1000215)

- Shipped on dry ice
- Store at -20°C



Visium HD 3' Spatial Gene Expression

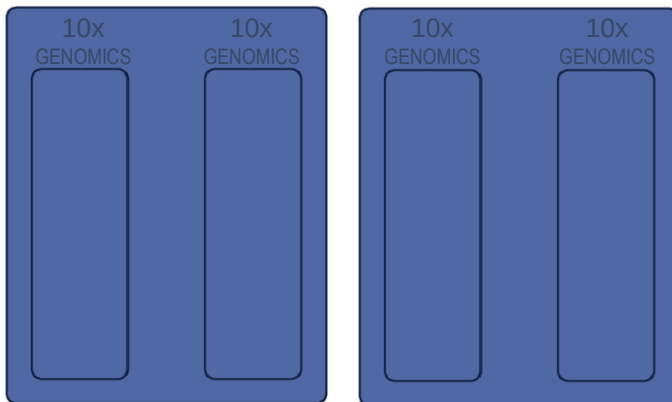
Required kits – Visium CytAssist Reagent Accessory Kit

Visium CytAssist Reagent Accessory Kit (PN-1000499)

- Low Profile Thermocycler Adapter (2)
- Magnetic Separator (1)
- Shipped at ambient
- Store at ambient



Low Profile (LP) Thermocycler Adapter



10x Magnetic Separator (A: 8-well or B: 4 x 8-well)

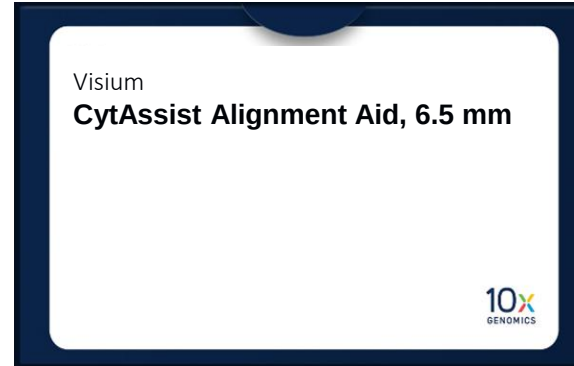


Visium HD 3' Spatial Gene Expression

Optional: Accessory Kit

Visium CytAssist Alignment Aid, 6.5 mm (PN-1000886)

- Alignment Aid, 6.5 mm (1)
- Shipped at ambient
- Store at ambient



Visium HD 3' Spatial Gene Expression



Refer to printing instructions to ensure printing to scale

Visium CytAssist Tissue Slide Alignment Quick Reference Card (QRC)

Use of Optional Alignment Aid, 6.5 mm

- **Annotate** the back of the tissue slide to **assist with alignment** onto the Visium CytAssist instrument.
- Consult **assay documentation** for the appropriate timing to use aid.
- **Practice** using the Aid with a stained tissue or mock-sample drawn on the slide is recommended.

Visium CytAssist
Instrument Alignment (Alignment Aid, 6.5 mm)
10x GENOMICS

The "Visium CytAssist Alignment Aid, 6.5 mm" (PN-1000886; available for purchase separately) is an optional accessory used to annotate the back of the tissue slide to assist with alignment onto the Visium CytAssist instrument. Consult specific assay documentation for the appropriate timing.

Overview

Aid is held with the text facing you

8 Mark Guides (trace these in step 3)

Square Window for centering tissue area of interest (DO NOT trace)

End Result (in step 4)

- Before use, verify tissue placement using the previous Tissue Slide Alignment pages
- Clean the Aid with 70% Isopropanol
- Marks on the back of the slide will not affect downstream imaging
- Not tested with Visium v2 assays

1 Hold the Alignment Aid in one hand **horizontally**, with the text oriented towards you. In the other hand, hold the tissue slide (tissue facing up) by the frosted end.

DO NOT touch tissue or coverslip

2 Position the slide on top of the aid, with the tissue area of interest centered on the square window. Move thumb to secure the slide against the aid.

hold slide/aid horizontal

slide must be parallel to aid; do not angle

3 Select a permanent ultrafine-tipped marker (dark color). Trace through the Mark Guides from the **BACK** of the Aid onto the back of the slide. **DO NOT** trace the window. Draw multiple passes for each mark. Allow ink to dry for ~5 min.

DO NOT tilt the slide/aid to view

4 Proceed through the protocol steps to **Tissue Slide Loading (onto the instrument)**. Match the drawn annotations on the back of the tissue slide to the alignment guides on the Tissue Slide Stage.

Align upper 11 mm marks first

Align 6.5 mm marks second

Lower 11 mm marks do not overlap (allows for rotating slide 180°)

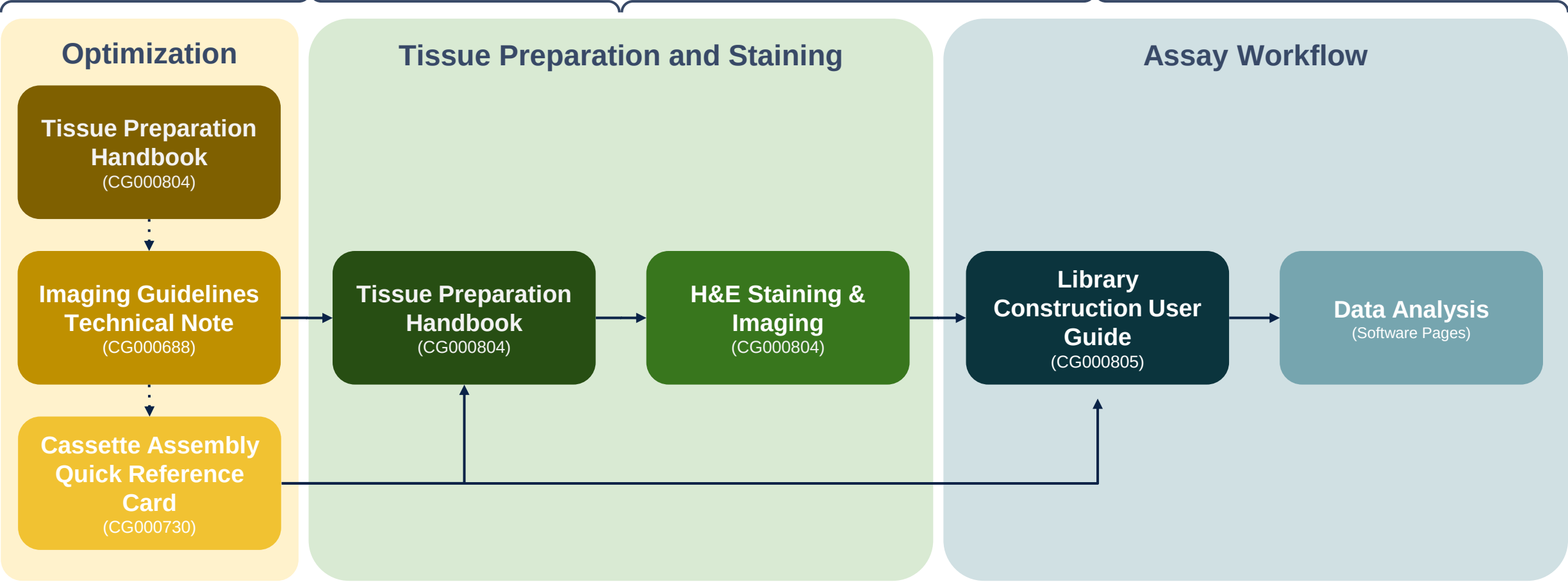
Visium CytAssist Quick Reference Cards – Accessory Kit | CG000548 | Rev E | 10xgenomics.com 6

Protocol Steps & Timing

General Resources for Visium HD 3' Assay Training

Before FAS Arrives

During FAS On-Site Training



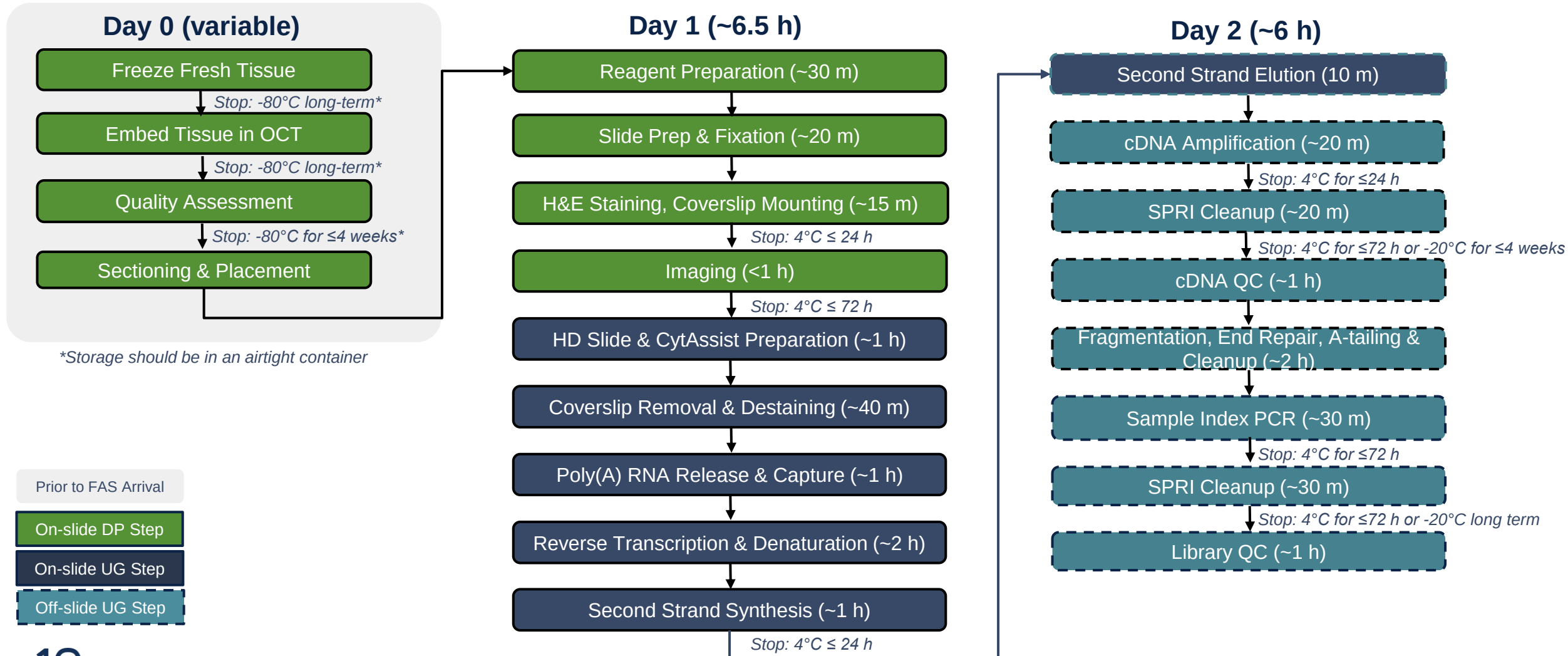
Depending on training plan and trainee expertise, FAS may not be on-site for the staining protocol

Visium HD 3' Spatial Gene Expression



Timing based on processing 2 tissue slides with a single Visium CytAssist instrument

Protocol steps and timing: Flexible 2-day workflow



Tissue Preparation Overview

Tested glass slides

PROTOCOL PLANNER | CG000803 Rev A1 | 10xgenomics.com

Visium HD 3' Spatial Gene Expression

Visium HD 3' Spatial Gene Expression Tested Slides

The following slides have been tested for use with the Visium CytAssist instrument. Diagrams for verifying that tissue sections are placed in the allowable area can be found in the Visium CytAssist Quick Reference Cards - Accessory Kit (Document CG000548). While the manufacturer's description of some slides may imply that they are compatible with 10x Genomics products, manufacturing tolerances may lead to dimensions that are incompatible with 10x Genomics products. Slide dimensions must be within 24.8 mm - 26.25 mm (width), 74.4 mm - 76.2 mm (length), and 0.96 - 1.07 mm (thickness).

Fresh Frozen Tissue - Recommended Slides

Description	Supplier	PN	Length (mm)	Width (mm)	Thickness (mm)	Important Considerations	Customer Notes
☐ Eprelia Shandon ColorFrost Plus	Fisher Scientific	67-762-14	75	25	1		
☐ Fisherbrand SuperFrost Plus	Fisher Scientific	12-550-15	75	25	1		
☐ Leica X-tra Clipped Corner (White color listed; other colors are acceptable)	Leica Biosystems	3800050CL	76.2	25.4	1		
☐ Leica Permaflex Plus (Pink color listed; other colors are acceptable)	Leica Biosystems	3800019	75.5	25.5	1		
☐ Schott Nexterion Slide H - 3D Hydrogel Coated Slides	Fisher Scientific	1800434	75.6	25	1	10x Genomics recommends placing tissues with large amounts of connective tissue or those prone to detachment on these slides to minimize risk of tissue detachment.	
☐ Sigma-Aldrich Poly Prep Slides	Millipore Sigma	P0425	75	25	1		
☐ Trajan Series 2 Adhesive Microscope Slide (Yellow color listed; other colors are acceptable)	Trajan	472042492	76	26	1		
☐ VWR SuperFrost Plus Micro Slide, Premium	VWR	48311-703	75	25	1		

Fresh Frozen Tissue - Not Recommended Slides

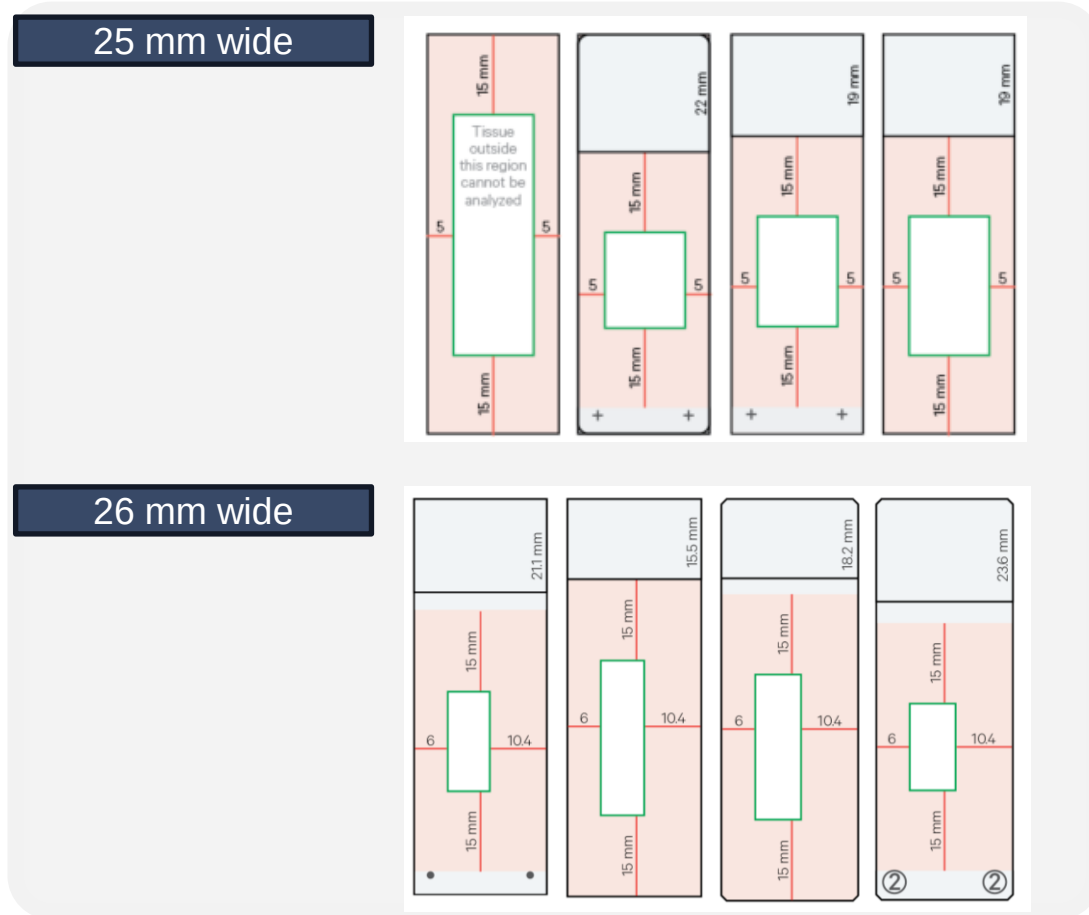
Description	Supplier	PN	Length (mm)	Width (mm)	Thickness (mm)	Important Considerations	Customer Notes
☐ Matsunami Platinum Pro	Matsunami	SPRO-05	75	26	1	Slide part not recommended. Partial detachment observed with a subset of tissue types tested.	

Utilize positively charged slides with compatible dimensions

- Tissue slide dimensions must be within specified width, length, and thickness to be compatible with the Visium CytAssist instrument
 - Minimum slide dimensions: 24.8 x 74.4 x .96 mm
 - Maximum slide dimensions: 26.25 x 76.2 x 1.07 mm
- Positively charged slides are designed to improve tissue adherence during on-slide workflow steps
 - We recommend placing tissue types prone to detachment such as those with large amounts of connective tissue and cellularly less dense on Schott Nexterion Slide H to minimize risk of detachment.
- For a detailed list of tested slide parts, review the Visium HD 3' Protocol Planner (CG000803).

Tissue Preparation Overview

Glass slide allowable area



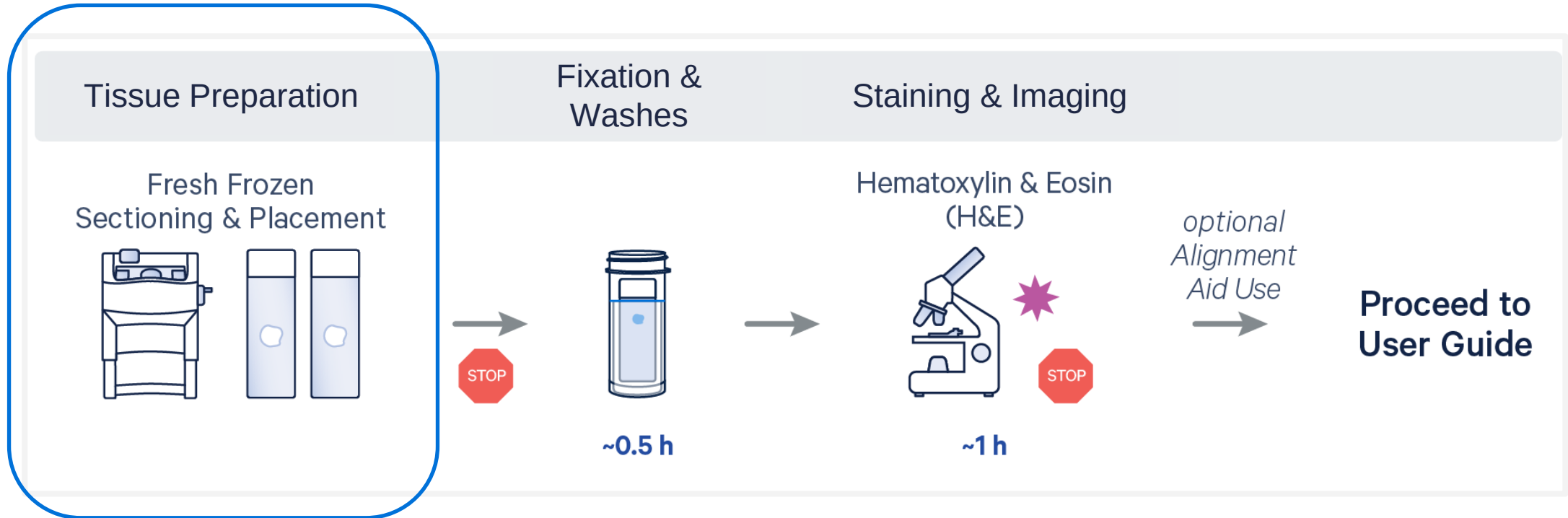
Place Tissue within Allowable Area

- Allowable Areas are properties of tissue slide types that vary based upon frosted area location and glass slide width
 - 26 mm wide slides have a smaller Allowable Area compared to 25 mm wide slides
- Tissue must be placed in the Allowable Area to be able to generate data
- This ensures sections are far enough away from frosted sections to not affect assay performance
 - Frosted sections include the opaque area of the slide as well as any etching on the top and/or bottom of the slide

Tissue Preparation

Sample Preparation Overview

Tissue Preparation



Tissue Preparation Overview

Compatible Tissue Features

Visium HD 3' Tested Tissues

Large breadth of tissues tested across multiple species

List of tissue types that were tested by 10x and found to perform well with the Visium HD 3' Spatial Gene Expression assay. Tissues tested in-house were prepared according to the [Visium HD 3' Tissue Preparation Handbook \(CG000804\)](#). Please note: sample preparation variability (such as sample handling, freezing, and embedding techniques), section thickness, tissue biology, and slide handling may impact assay performance. For additional information on Tested Slides and recommendations, please refer to the [Visium HD 3' Spatial Gene Expression Protocol Planner \(CG000803\)](#). Visium HD 3' datasets can be found [here](#).

Species	Tissue	Primary diagnosis
Canine	Brain	Healthy, Cerebral Cortex
Chicken	Ovary	Healthy
Chicken	Heart	Healthy
Cyno	Liver	Healthy
Human	Brain	Healthy, Amygdala
Human	Breast	Cancer, Infiltrating Ductal Carcinoma (IDC)
Human	Breast	Cancer, Ductal Carcinoma In Situ (DCIS)
Human	Colon	Cancer, Invasive Adenocarcinoma
Human	Liver	Portal Inflammation
Human	Liver	Cancer, Hepatocellular Carcinoma
Human	Liver	Portal Fibrosis
Human	Lung	Cancer, Papillary Adenocarcinoma
Human	Lymph Node	Reactive Follicular Hyperplasia
Human	Ovary	Cancer, Clear Cell Adenocarcinoma
Human	Pancreas	Cancer, Ductal Adenocarcinoma
Human	Skin	Cancer, Primary Dermal Melanoma
Human	Spleen	Healthy
Human	Tonsil	Hyperplasia
Mouse	Brain	Healthy
Mouse	Embryo E15	Healthy
Mouse	Heart	Healthy
Mouse	Kidney	Healthy
Mouse	Lung	Healthy
Mouse	Pup P1	Healthy
Mouse	Small Intestine	Healthy
Mouse	Spleen	Healthy
Mouse	Thymus	Healthy
Pig	Liver	Healthy
Rat	Liver	Healthy
Rat	Kidney	Healthy
Rhesus	Liver	Healthy
Rhesus	Kidney	Healthy
Xenograft	Mouse/Human	Cancer, Human Colorectal Carcinoma cell line (HCT116) grown in NSG mouse
Xenograft	Rat/Human	Cancer, Human Triple Negative Breast Cancer cell line (MDA-MB-231) grown in rat
Zebrafish	Head	Healthy

Challenging Tissues

List of tissue types tested by 10x with the Visium HD 3' Spatial Gene Expression assay that were found to not perform well following HD 3' protocols. Sample preparation and/or workflow modifications may be required with these tissue types.

Species	Tissue	Primary diagnosis
Human	Lung	Healthy



Visit the 10x Genomics Support Site for a comprehensive list of tissues tested

- Most **Fresh frozen; OCT embedded (FF)** tissue blocks
 - Alternative freezing or embedding protocols are not currently supported.
- Tissue section **thickness of 10 µm**
- RNA Quality Assessment
 - **RIN ≥ 7** is recommended
- Available on the 10x Genomics Support Site
 - **Tested Tissue List**
 - **Public datasets**

Tissue Preparation Overview

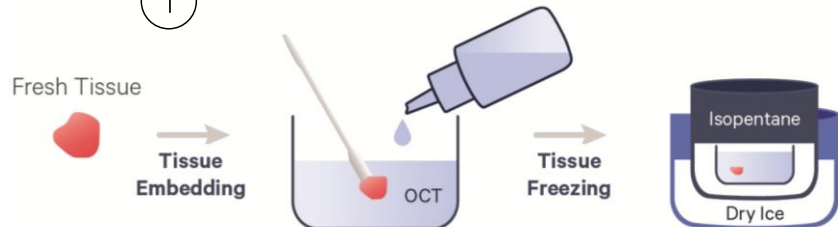
Tissue Slide preparation



See documentation for details on tissue preparation

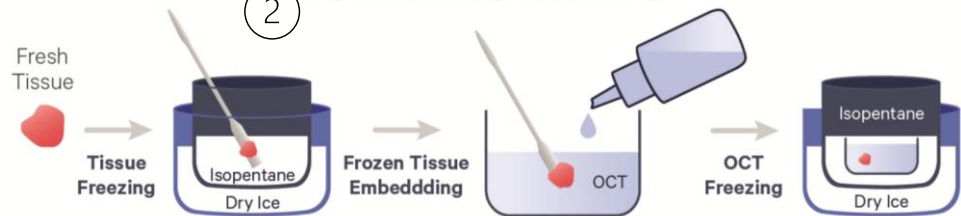
Freezing & Embedding – 2 methods

1 Simultaneous Freezing & Embedding



 **-80°C for long term storage**

2 Separate Freezing & Embedding

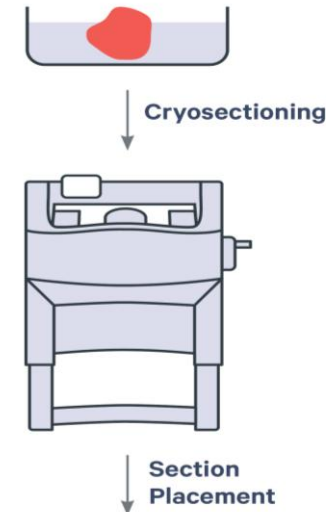


 **-80°C for long term storage**



DO NOT freeze tissues directly in liquid nitrogen.

Cryosectioning & Tissue Section Placement



 **-80°C for up to 4 weeks**

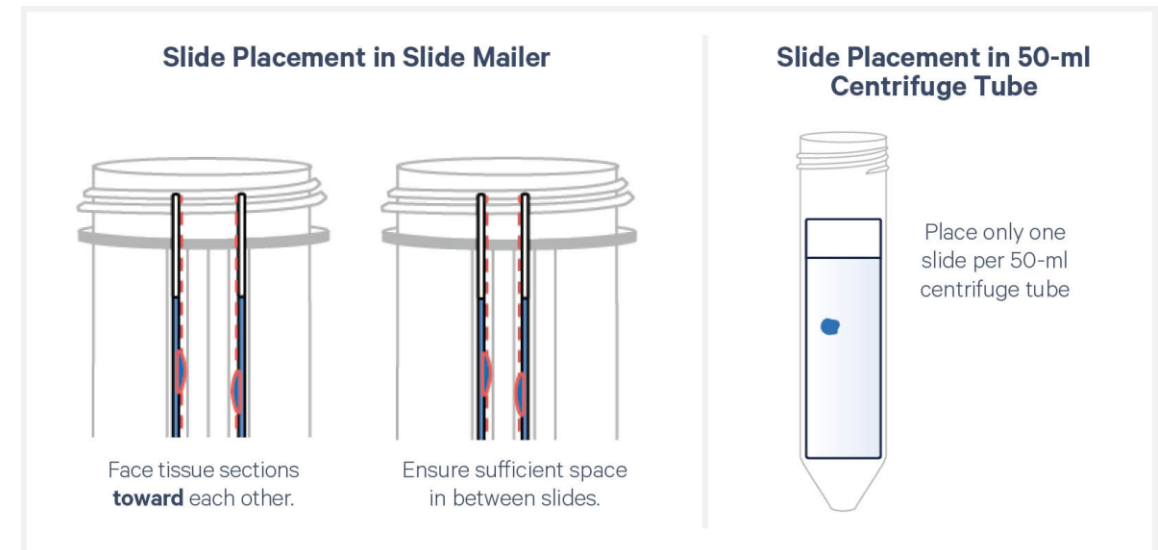
Tissue Preparation – Tissue Slide Shipping



Temperature fluctuations may result in RNA degradation.

Tissue Slide Shipping

- **Place** slides in pre-chilled slide mailer sitting in dry ice.
 - **Ensure** sufficient space between slides.
 - **Ensure** side(s) with tissue face inward.
 - 50 ml centrifuge tube alternative, place **one slide per tube**.
- **Place** mailer in a tightly sealed secondary container.
- **Ship** samples overnight in dry ice.
 - Ensure sufficient dry ice to account for transit and delivery times.
- **Observe** local institution and delivery service instructions for dry ice shipments.

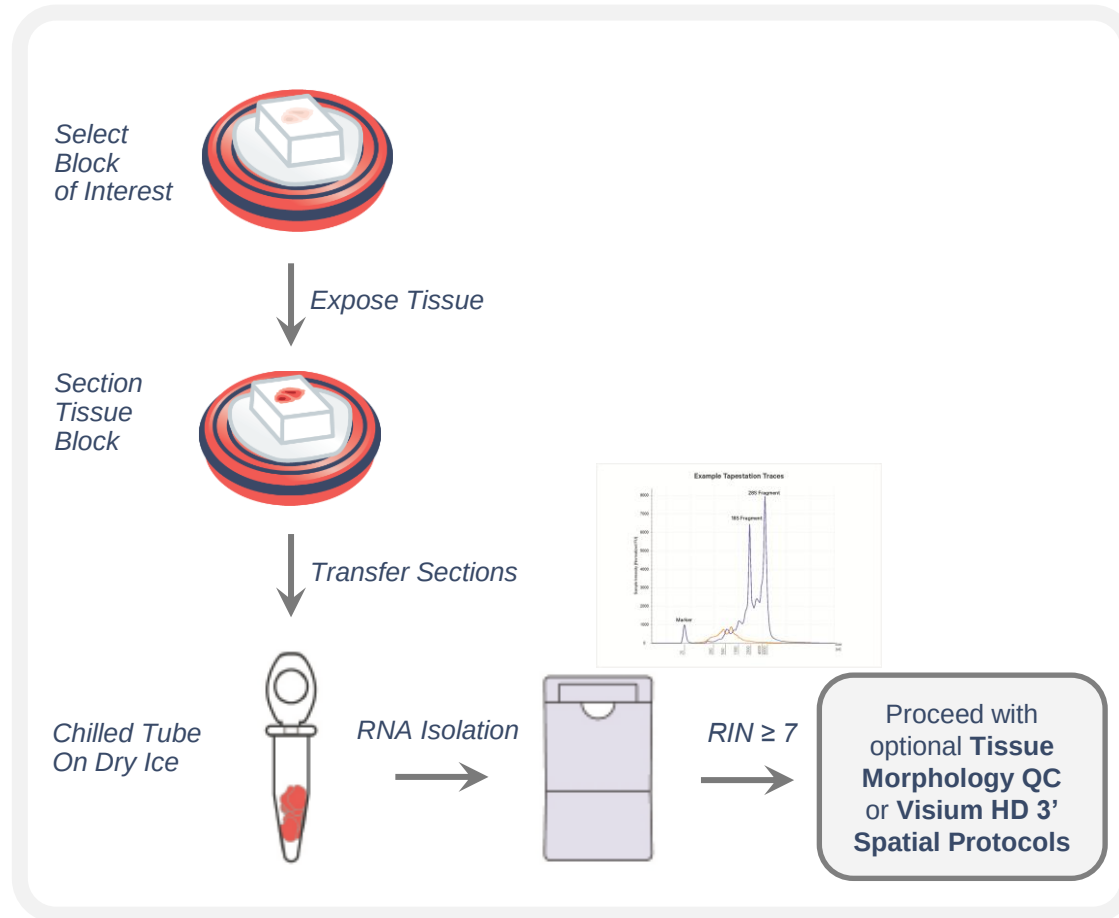


Tissue Preparation - QC

RNA Quality Assessment



See documentation for details on tissue preparation



RNA Quality Assessment

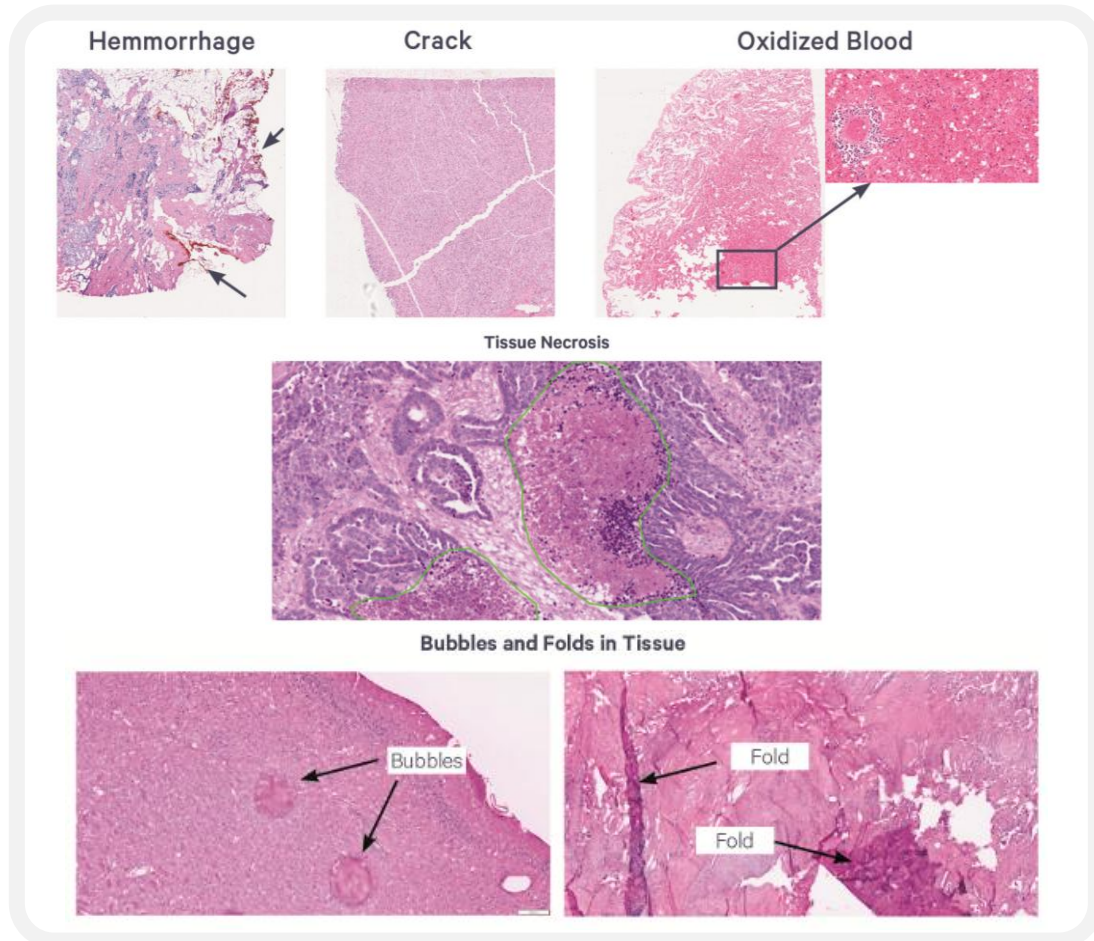
- RIN is an algorithm for assigning integrity values (ranging 1-10) to RNA measurements
- Collect **20 - 30 mg** of tissue sections inside a chilled 1.5 ml tube sitting on dry ice
- Perform **RNA extraction** by following manufacturer's instructions
- Using a Nanodrop or Qubit, **measure RNA concentration**
 - Determine the appropriate dilution for RIN evaluation
- Store RNA at -80°C or proceed to **RIN evaluation**
- Protocol was optimized using samples with **RIN ≥ 7**
 - Samples with $\text{RIN} \geq 7$ are more likely to generate higher quality data
- See Visium HD 3' Fresh Frozen Tissue Preparation Handbook (CG000804) for details

Tissue Preparation - QC

Optional Tissue Morphology Assessment



See documentation for details on tissue preparation



H&E Tissue Morphology Assessment

- Ensure the tissue section is free from tissue processing and sectioning artifacts such as due to:
 - Improper snap freezing
 - Squeeze/crush artifacts
 - Ice crystals
 - Hemorrhaging
- Sections affected by the above artifacts are unlikely to generate good data in the affected regions.



*This is not an exhaustive list of
third-party items – see the
Protocol Planner*

Key Third-party Items

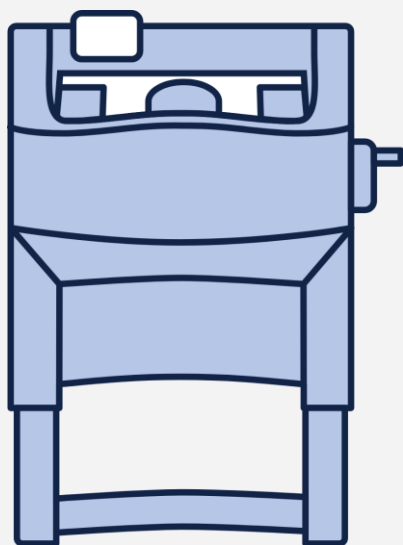
Visium HD 3' Spatial Gene Expression



See Protocol Planner at ship for a comprehensive list of third-party items

Key third-party reagents, consumables, and tools

Handbook: Tissue Preparation



Reagents, Consumables & Tools

(Freezing, Embedding, Sectioning, Tissue Placement, and QC)

- ✓ Isopentane
- ✓ Dry Ice
- ✓ Optimal Cutting Temperature Compound, OCT
- ✓ Disposable OCT molds
- ✓ Brushes or probes
- ✓ Compatible glass slides*
- ✓ RNA extraction kit and associated reagents
- ✓ High Sensitivity RNA Gel Electrophoresis Kit
- ✓ *Liquid Nitrogen (optional)*
- ✓ *H&E Staining dyes (optional)*

**See protocol planner for tested slides & considerations*

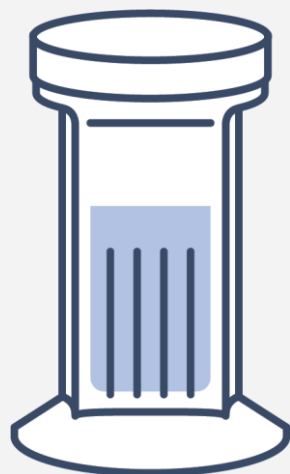
Visium HD 3' Spatial Gene Expression



See Protocol Planner at ship for a comprehensive list of third-party items

Key third-party reagents, consumables, and tools

Handbook: H&E Staining



Reagents & Consumables

(H&E Staining)

- ✓ Acetone
- ✓ Methanol
- ✓ Nuclease-free water
- ✓ 1X or 10X PBS, pH 7.4
- ✓ 0.1N HCl
- ✓ Gill II Hematoxylin
- ✓ Alcoholic Eosin
- ✓ Bluing Buffer
- ✓ 100% polyester lint-free wipes
- ✓ Glycerol
- ✓ Coverslips
- ✓ *Ultra Fine Point Permanent Marker (optional)*

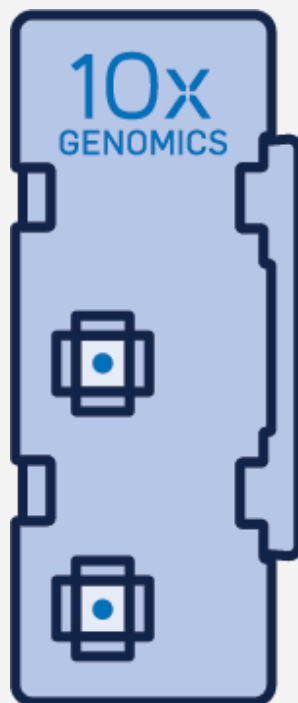
Visium HD 3' Spatial Gene Expression



See Protocol Planner at ship for a comprehensive list of third-party items

Key third-party reagents, consumables, and tools

User Guide: Library Construction



Reagents & Consumables

- ✓ 1X Low TE Buffer, pH 8.0
- ✓ 1M Tris-HCl, pH 7.0
- ✓ Nuclease Free Water
- ✓ 1X or 10X PBS, pH 7.4
- ✓ 20X SSC Buffer
- ✓ Buffer EB
- ✓ 8M KOH
- ✓ SPRIselect reagent
- ✓ 100% polyester lint-free wipes
- ✓ High Sensitivity DNA Gel Electrophoresis Kit

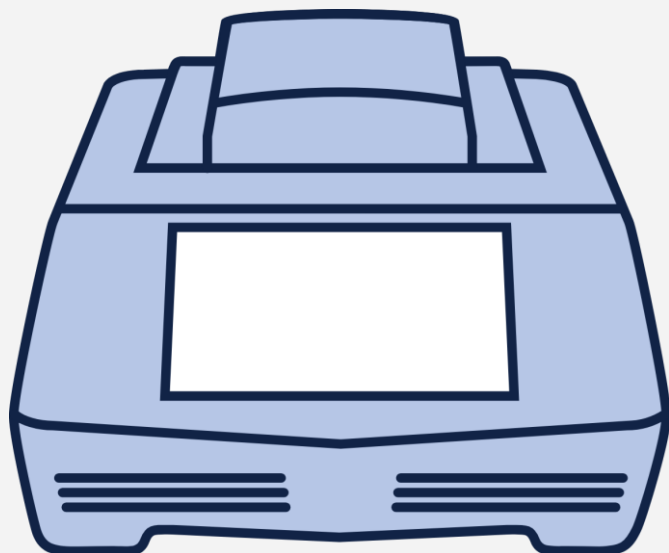
Visium HD 3' Spatial Gene Expression



See Protocol Planner at ship for a comprehensive list of third-party items

Key third-party reagents, consumables, and tools

General Equipment



Equipment

- ✓ 2 Temperature controlled instruments*: Heat Block, Bead Bath or Thermomixer
- ✓ Cryostat and associated parts
- ✓ 1 Supported thermal cycler tested by 10x[†] for processing 2 Tissue Slides = 1 HD Slide
- ✓ Imaging System[‡]
- ✓ Qubit or Nanodrop
- ✓ Gel Electrophoresis QC System

**Water baths are not recommended.*

[†] See protocol planner for tested cyclers & considerations before performing the assay.

[‡] Refer to HD Imaging Guidelines TN for imaging system requirements.

Transition: Visium HD to Visium HD 3'

New third-party equipment, reagents, consumables, and tools



See Protocol Planner at ship for a comprehensive list of third-party items

Equipment



2 Temperature controlled instruments: Heat Block, Bead Bath or Thermomixer*

1. 37°C
2. 65°C

**Water bath not recommended*

Reagents



- Methanol
- Acetone
- 1X Low TE Buffer, pH 8.0
- 1M Tris-HCl, pH 7.0
- 70% Isopropanol or 70% Ethanol
- RNase Decontamination Solution

Consumables & Tools



- 15 ml centrifuge tubes
- 2 ml microcentrifuge tubes
- *Ultra Fine Point Permanent Marker, dark color (optional)*

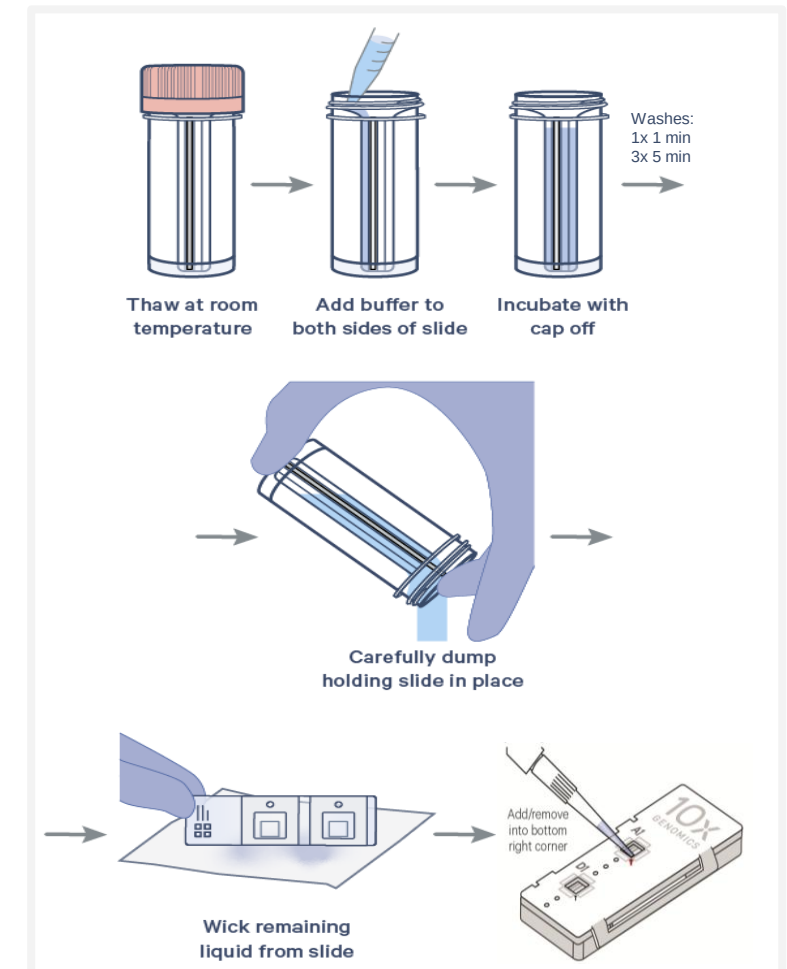
Assay Overview

User Guide Steps

Step 1: Visium HD Slide & Instrument Preparation

Visium HD Slide Preparation

1. **Remove** mylar bag containing HD slide from -80°C
2. **Thaw** slide mailer sitting upright with cap on at RT (30 m - 3 h)
3. **Remove cap** from mailer
4. **Wash** slide in mailer, ensuring capture areas are completely submerged
5. **Hold** slide securely while discarding buffer
6. **Wick** remaining liquid using a lint-free laboratory wipe
7. **Inspect** to ensure slide is free of debris
 - Immerse slide quickly in and out of buffer in mailer to remove
8. **Record** the Slide Serial Number
9. **Assemble** in Visium Cassette S3
10. **Wash** cassette wells with buffer, removing ALL liquid
11. **Add** Pre-equilibration Mix and seal cassette
 - DO NOT exceed 1 hour before initiating CytAssist run



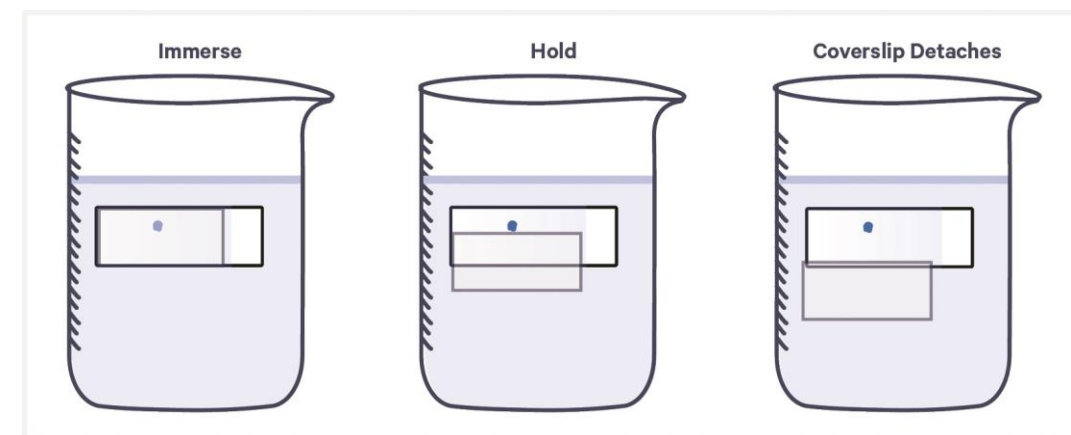
Step 2: Coverslip Removal & Destaining

Tissue Slide Coverslip Removal

1. Ensure **beakers** have been **cleaned** per recommendations
2. **Remove coverslip** by incubating slide in water while holding sideways until slowly separates away
3. **Immerse** slide 30x to remove residual mounting media



- *DO NOT* move slide up and down or shake forcibly or manually move coverslip.
- Ensure mounting media is fully removed. Residual media will likely affect downstream steps.



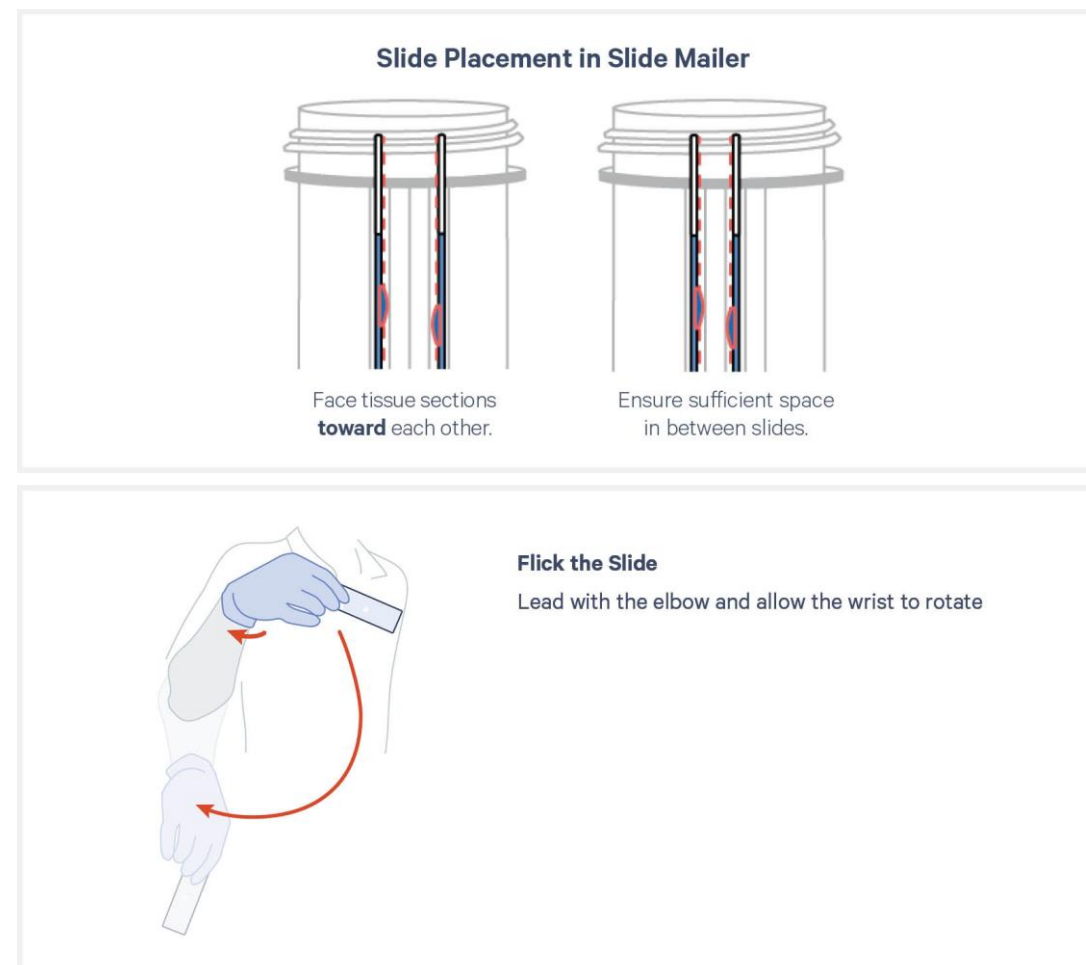
Step 2: Coverslip Removal & Destaining

Tissue Slide Destaining

1. **Retrieve tissue slides** from 4°C
2. **Remove Hematoxylin** from tissue
3. **Start preparation** of Permeabilization Mix
4. **Immerse & incubate** slides in PBS
5. **Flick** the tissue slides until all buffer has been dislodged from surface of tissue
6. **Place** slides on lint-free wipe



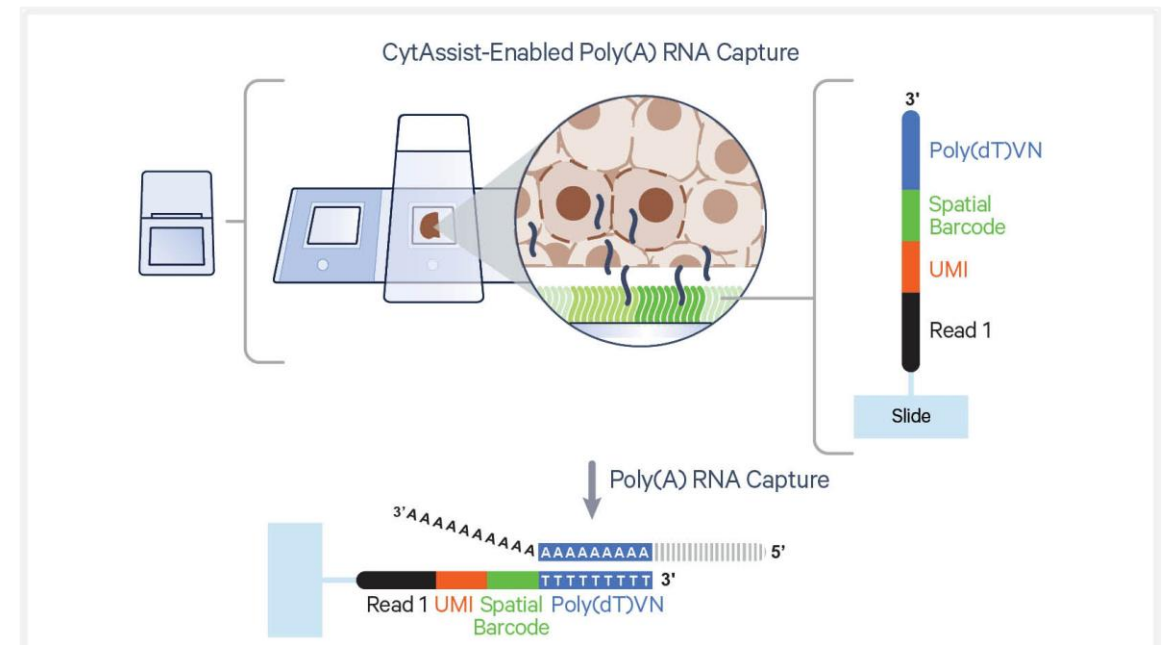
- Ensure tissue sections are completely submerged.
- DO NOT make complete Permeabilization Mix.
- DO NOT exceed 5 min in PBS buffer after destaining.



Step 3: Poly(A) RNA Capture, RT, & Denaturation

CytAssist-Enabled Poly(A) RNA Release & Capture

- Visium CytAssist instrument running firmware **v2.0.0 or higher**:
 - During the run, the tissue is permeabilized releasing poly(A) RNA
 - The released poly(A) RNA is captured by the oligo poly(dT)VN sequence on the Visium HD slide directly below
 - It is normal for tissue to remain on the Tissue Slide after this step



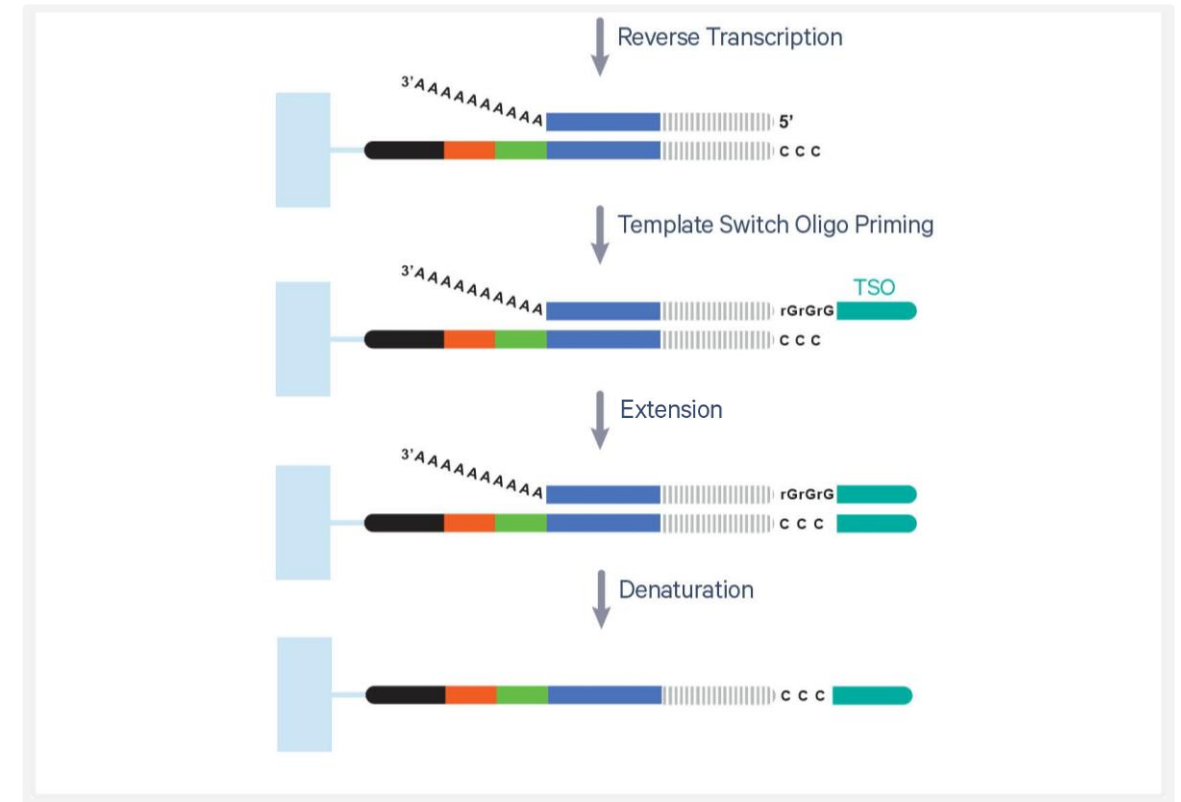
Step 3: Poly(A) RNA Capture, RT, & Denaturation

Reverse Transcription & Denaturation

- RT Master Mix is added to the Visium HD Slide. This produces spatially barcoded, full-length cDNA from poly(A) RNA captured on the slide.
- Two rounds of Reverse Transcription are required
- Captured poly(A) RNA is denatured and washed away, leaving first strand cDNA attached to HD slide via barcoded oligos



Ensure KOH is prepared fresh at 0.08M



Step 4: Second Strand Synthesis & Elution

- Second strand mix is added to first strand cDNA attached to the HD slide to generate double stranded cDNA

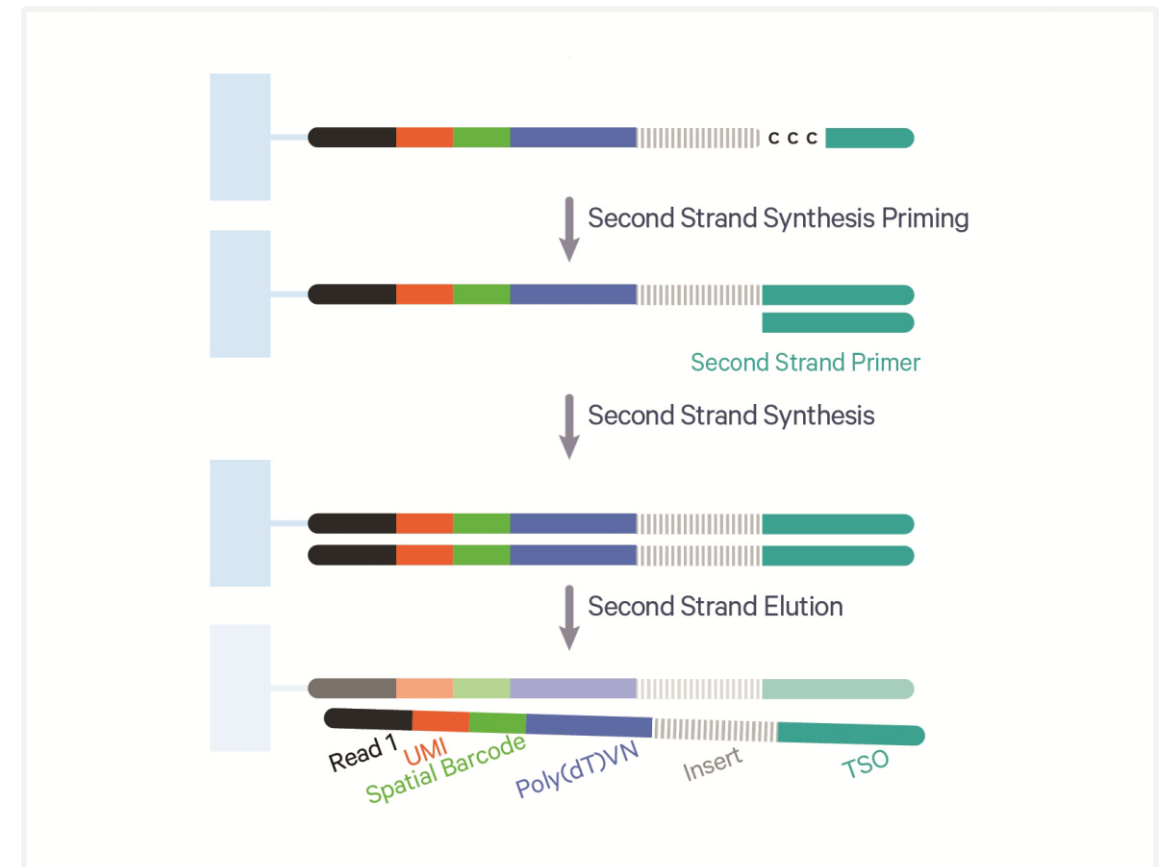


4 °C for ≤ 24 hours

- Newly synthesized second strand cDNA is eluted off the slide, collected in an 8-well strip tube and neutralized



- Ensure KOH is prepared fresh at 0.08M
- Ensure 1M Tris-HCl pH 7.0 is used for neutralization



Step 5: cDNA Amplification & SPRIselect

- Spatially barcoded, single stranded cDNA is amplified with a fixed number of cycles to generate additional product

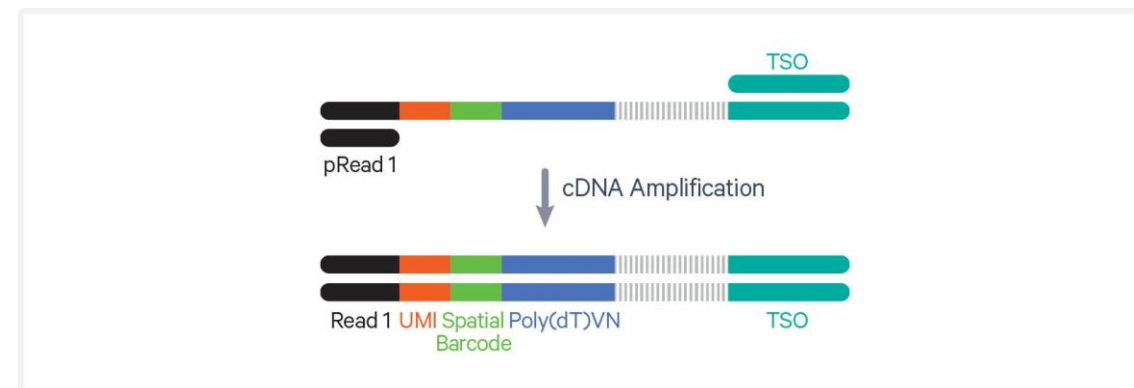


4°C for ≤ 24 hours

- Amplified cDNA is then cleaned up with single sided SPRIselect



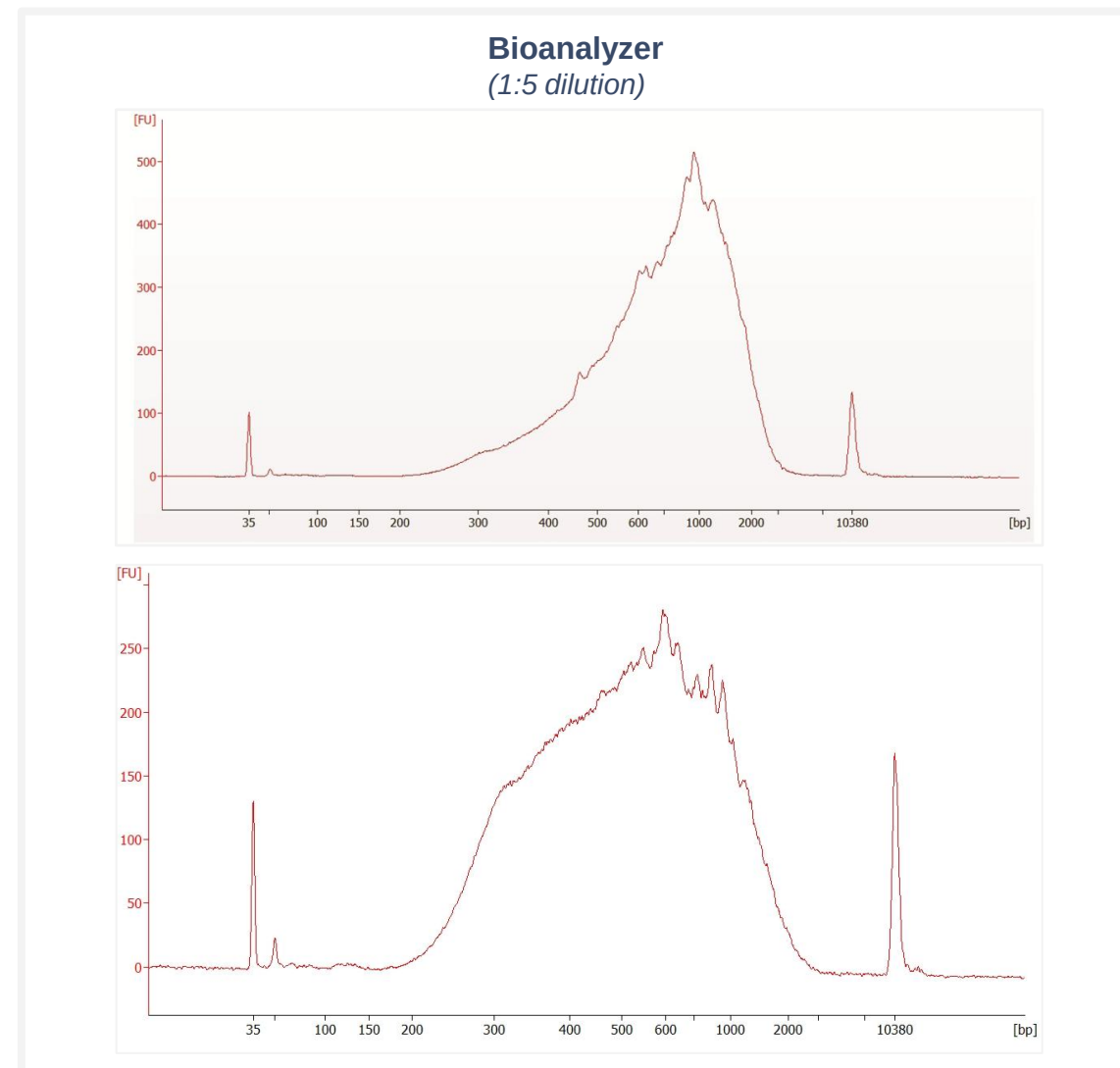
4°C for ≤ 72 hours, -20°C for ≤ 4 weeks



Step 5: cDNA Amplification & SPRIselect

cDNA QC & Quantification

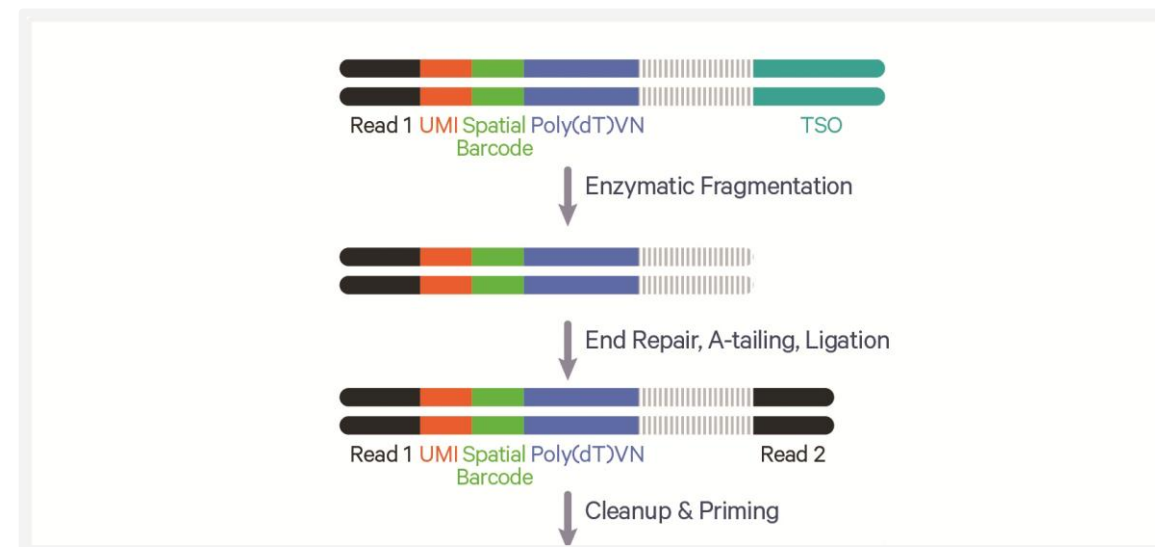
- cDNA is diluted 1:5, may not be optimal for all samples
 - Ensure concentration is within the specification range of electrophoresis assay kit
- cDNA concentration and average size is determined using a Bioanalyzer or similar automated electrophoresis instrument
 - cDNA average size expected ~500-1000bp
 - cDNA yield calculated to determine number of cycles to use during Sample Index PCR step



Step 6: Library Construction

Fragmentation, End Repair, & A-tailing

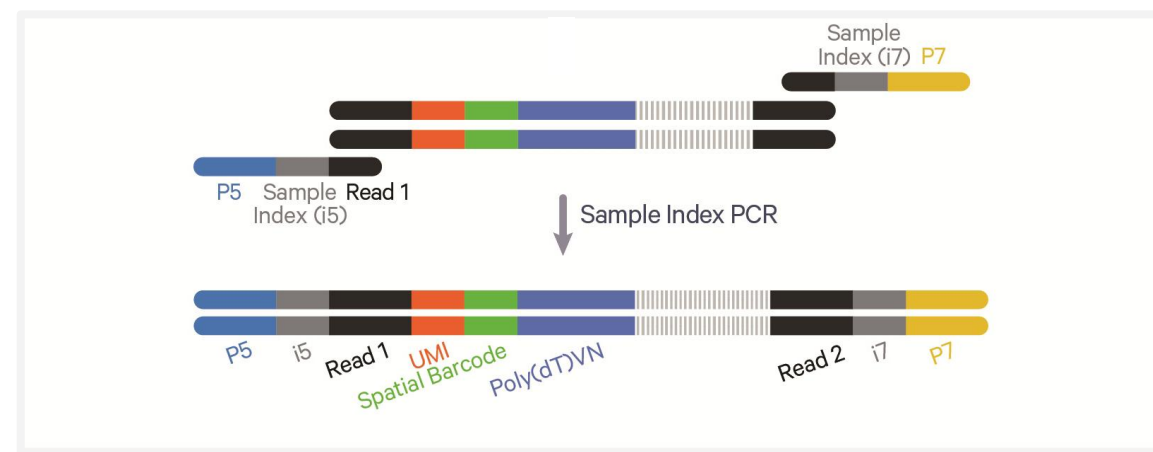
- Enzymatic Fragmentation followed by double sided SPRI are performed to optimize cDNA amplicon size
- 25% of cDNA is used as input
- End Repair & A-tailing prepare the 3' end
- Adaptor ligation adds a partial Read 2 sequence and is followed by a single sided SPRI



Step 6: Library Construction

Sample Index PCR (SI-PCR)

- Spatially barcoded cDNA are amplified to add Sequencing Adapters (P5, P7) and Sample indexes (i5, i7)
 - Requires Dual Index Plate TT Set A containing 1 unique i5 and i7 index per well
 - PCR cycle number based on fragmentation reaction cDNA input mass (25% of cDNA yield)
 - STOP 4°C for ≤ 72 hours
- PCR products then undergo doubled sided SPRIselect cleanup
 - STOP 4°C for ≤ 72 hours, -20°C for long term storage

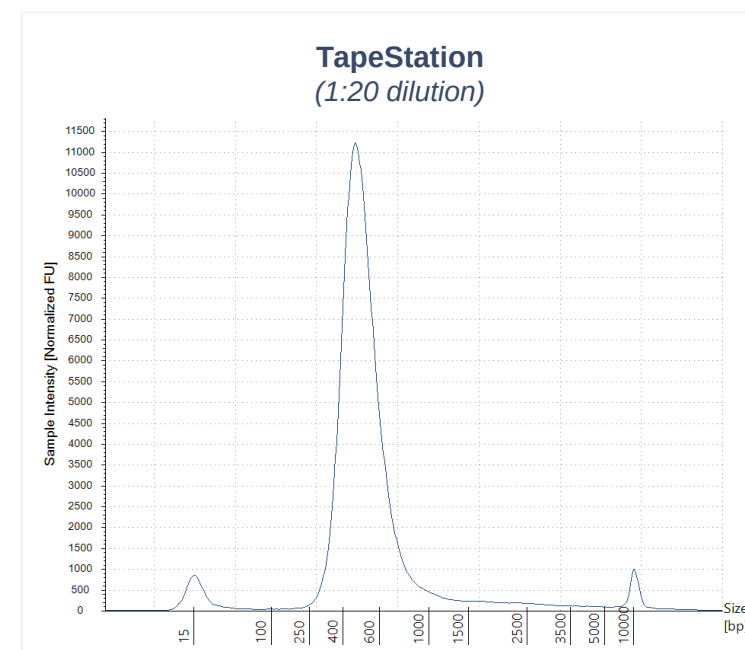
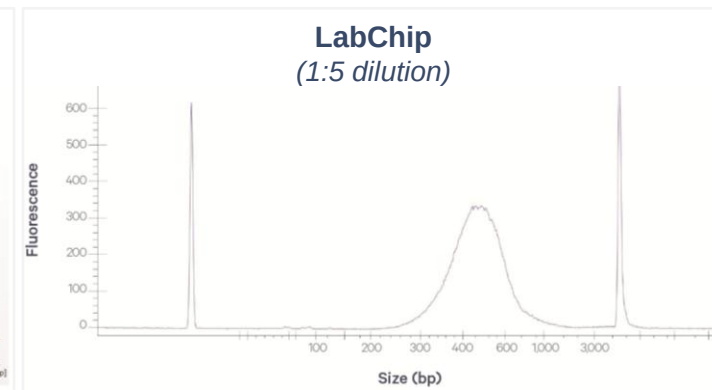
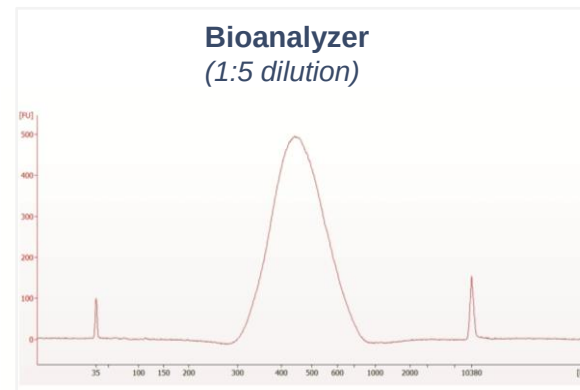


cDNA Input	Total Cycles
1-10 ng	15
10-50 ng	14
50-100 ng	13
100-250 ng	12
250-350 ng	11
350-600 ng	10
600-800 ng	9
800-1,100 ng	8
1,100-1,300 ng	7
1,300-1,500 ng	6
>1,500 ng	5

Step 6: Library Construction

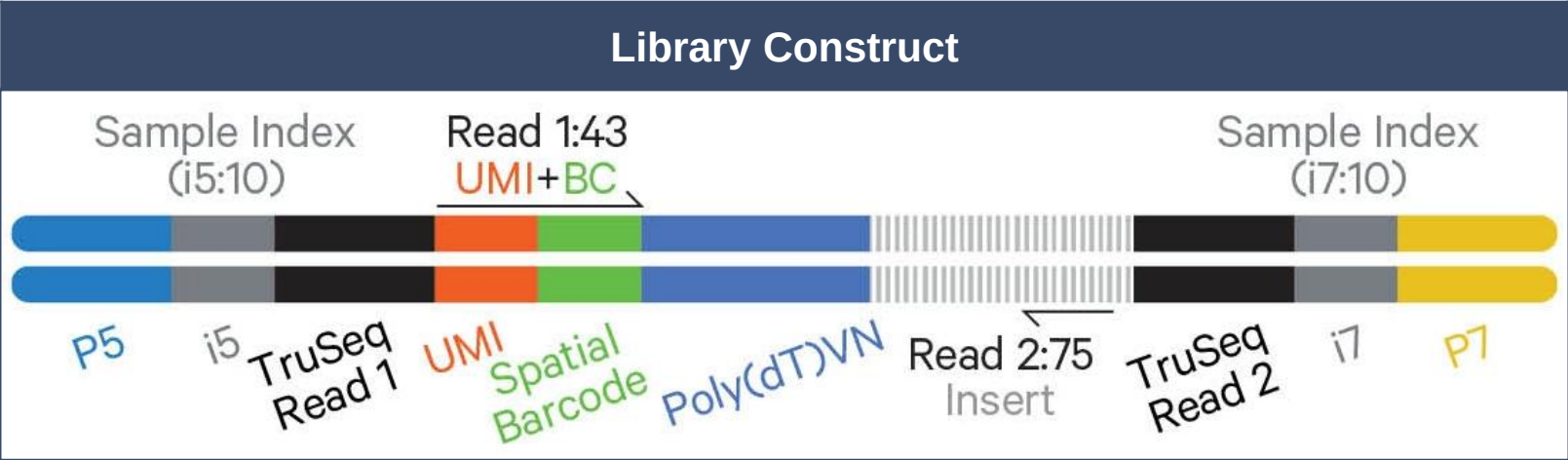
Post-Library Construction QC

- Library is diluted 1:5, may not be optimal for all samples
 - Ensure concentration is within the specification range of electrophoresis assay kit
- Quality and size is assessed using a Bioanalyzer or similar automated electrophoresis instrument
 - Expect sharp peak ~450 bp in size
 - Additional peaks can indicate inadequate dilution and/or poor SPRIselect cleanup
- KAPA qPCR library quantification is performed to calculate sequenceable molecules containing P5 and P7 adaptors



Visium HD 3' Spatial Gene Expression

Sequencing Configuration



Recommended Sequencing Configuration & Parameters				
Paired-end, Dual Index Sequencing				
	Read 1	i7	i5	Read 2
Purpose	UMI + Spatial Barcode	Sample Index	Sample Index	Insert
Cycles	43	10	10	75

*Do not recommend pooling with other libraries

Visium HD 3' Spatial Gene Expression

Sequencing Depth Calculation

- Estimate fraction of capture area covered by tissue by visual observation
 - How to calculate sequencing depth:
 - Minimum sequencing depth = (550 million read pairs * Fraction of capture area covered)
Example with 60% coverage:
 $550,000,000 * 0.60 = 275 \text{ million read pairs}$
- Alternatively, this can be determined using the manual workflow in Loupe Browser (v8.0 or higher)

Minimum Sequencing Depth Recommendation

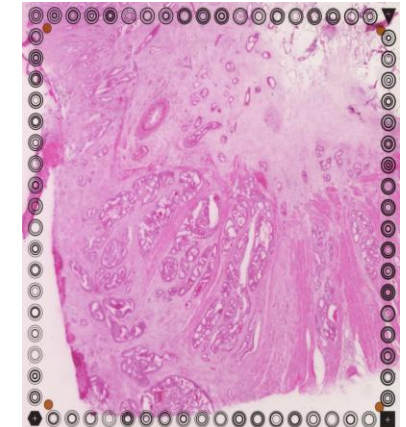
550 million read pairs per fully covered capture area
*(550 million read pairs) * (fraction of capture area covered by tissue)*

~50% tissue coverage
Mouse Brain



$550M * 0.50 = 275M \text{ read pairs}$

~95% Tissue coverage
Human Colon Cancer



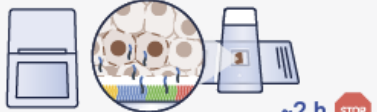
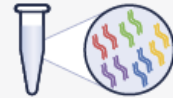
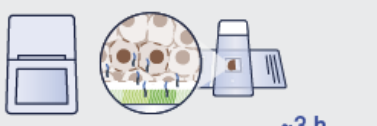
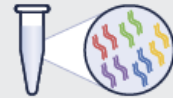
$550M * 0.95 = 523M \text{ read pairs}$

Transition: Visium HD to Visium HD 3'

	Visium HD	Visium HD 3'
Sample Preservation	Fresh Frozen, OCT embedded	
Species Compatibility	Mouse and Human	Species Agnostic
Capture Area Size	6.5 mm	6.5 mm
Studies Enabled	Whole Transcriptome, Panel	Unbiased Whole Transcriptome (3')
Transcript Detection	Probe-based Capture with v2 Probe Panels	Poly(A) RNA Capture
Transcript Method	CytAssist-Enabled	
Tissue Staining	H&E or Immunofluorescence	H&E
Tested Tissues	13	>30
Sample Quality	RIN of 4 or greater	RIN of 7 or greater
Sample Input	Fresh cut sections on 25- or 26-mm wide glass slides	
Section Thickness	10 - 20 µm	10 µm
Minimum Sequencing Depth Recommendation	275M Read Pairs per fully covered capture area with tissue	550M Read Pairs per fully covered capture area with tissue
Sequencing Configuration	R1 - 43, i5 - 10, i7 - 10, R2 - 50	R1 - 43, i5 - 10, i7 - 10, R2 - 75

Transition: Visium HD to Visium HD 3'

Workflow Steps & Timing

	Sample Preparation	Staining & Imaging	Probe Hybridization & Ligation	Slide Preparation	Permeabilization & Barcoding	Library Construction
Visium HD	FF: Freezing, Embedding, Sectioning & Placement 	Fixation  ~0.5 h H&E Staining, Imaging, Destaining & Permeabilization OR Permeabilization, IF Staining & Imaging  >1 h STOP >1 h	Whole Transcriptome Probe-Mediated mRNA Detection  Overnight <2 h STOP	Thawing & Washing on Visium HD slide  <1 h	CytAssist-Enabled Probe Release & Extension  ~2 h STOP	Elution, Pre-amplification, & Cleanup  ~1.5 h STOP qPCR, SI-PCR & QC  ~3 h STOP
Visium HD 3'		Fixation  ~0.5 h H&E Staining & Imaging  >1 h STOP		Thawing & Washing on Visium HD slide Destaining on Tissue Slide  <1 h	CytAssist-Enabled poly(A) RNA Capture, Reverse Transcription & Denaturation  ~3 h	Second Strand Synthesis, Elution, cDNA amplification, & Cleanup  ~1.5 h STOP Fragmentation, End Repair, A-Tailing, Adaptor Ligation, SI-PCR, Cleanup & QC  ~3 h STOP

2-3 days from tissue sections on slides to sequencing ready libraries